A Smart Food Ordering and Waste Reduction System for Train Services: A Comparative Mixed-Method Analysis

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***Abstract*—Railway food services operate under constraints such as variable passenger demand, limited on-board storage, constrained preparation facilities, and tight delivery windows defined by train schedules. These conditions lead to frequent overproduction, stock-outs, and inconsistent food quality, which in turn create food waste and passenger dissatisfaction. In parallel, a growing body of research has examined digital food ordering platforms, institutional food service forecasting, IoT-based quality monitoring, blockchain traceability, and circular economy initiatives in stationary and commercial contexts. This paper reviews and compares such work using a mixed-method approach to derive evidence-based insights for designing a smart food ordering and waste reduction system for train services, without implementing a new system in practice.**

***Index Terms*—railway catering, food waste reduction, demand forecasting, digital traceability, mixed-method analysis**

I. Introduction

Railway food services operate under constraints such as variable passenger demand, limited on-board storage, constrained preparation facilities, and tight delivery windows defined by train schedules. These conditions lead to frequent overproduction, stock-outs, and inconsistent food quality, which in turn create food waste and passenger dissatisfaction. Traditional processes also provide limited visibility into inventory movement, cold-chain conditions, and supplier performance along the route.

In parallel, a growing body of research has examined digital food ordering platforms, institutional food service forecasting, IoT-based quality monitoring, blockchain traceability, and circular economy initiatives in stationary and commercial contexts. These studies demonstrate that mobile ordering, IoT monitoring, machine learning (ML) forecasting, and digital traceability can improve demand prediction, product quality, and transparency. However, most of this evidence comes from campuses, institutional kitchens, agricultural supply chains, or online delivery platforms, not from railway environments.

This article therefore examines how existing systems and technologies, studied in other domains, could inform the design and evaluation of a smart food ordering and waste reduction system for train services. The focus is strictly on analysing and comparing current research contributions; no new system is implemented in practice.

II. Problem Statement

The railway food service sector faces several well-recognised challenges:

- *Unpredictable demand*: passenger numbers and ordering behaviour vary by route, time, and station, making accurate production planning difficult and leading to overproduction or shortages.
- *Food waste and cost*: overproduction, limited storage, and time-bound delivery windows cause surplus food that cannot be reused safely, increasing waste and operating cost.
- *Limited transparency*: existing railway catering processes provide little end-to-end visibility on where, when, and under what conditions food items move from suppliers to trains and passengers.
- *Inconsistent passenger experience*: passengers frequently report issues with availability, freshness, and reliability of on-board food services, affecting satisfaction and trust.

At the same time, the current research landscape is fragmented. Many studies address individual aspects—such as digital traceability, ML-based demand forecasting, consumer behaviour in online food delivery, IoT-based cold chain monitoring, smart kitchen waste tracking, and circular economy models—but rarely consider railway-specific constraints or integrated evaluations.

The problem addressed in this paper is therefore not to build a new system, but to analyse and synthesise existing research on smart food ordering, waste reduction, and digital supply chain technologies. The goal is to derive evidence-based insights and comparative benchmarks that can inform future design and evaluation of railway-focused solutions.

III. Research Methodology

A. Mixed-Method Comparative Design

This study employs a mixed-method research design that combines quantitative and qualitative analysis. No new technical system is deployed; instead, existing implementations and empirical findings are reviewed and compared.

The quantitative component focuses on reported performance metrics from the selected studies, such as waste reduction percentages, forecasting accuracy, service-level indicators, cost impacts, and environmental outcomes. The qualitative

component analyses how these systems were designed and implemented, the challenges encountered, and the contextual factors that influenced success or failure. Together, these perspectives support a comparative mixed-method evaluation of current digital food service systems and their relevance to railway food services.

B. Data Collection

Data are collected entirely from secondary sources described in the ten selected articles and reports. For each study, the following information is extracted:

- *Context and scope*: sector (institutional kitchens, agricultural supply chains, online food delivery, cold chain logistics, circular economy projects), scale, and timeframe.
- *Technologies and interventions*: use of ML-based demand forecasting, IoT monitoring, blockchain or other traceability tools, smart kitchen systems, and organizational change strategies.
- *Quantitative results*: reported waste reduction ranges, forecasting error metrics, response times, consumer satisfaction indicators, cost changes, and emission reductions where available.
- *Qualitative findings*: implementation challenges, organizational barriers, user behaviour, equity considerations, and identified research gaps.

This structured data extraction mirrors the summary tables used in the base research and forms the basis for both descriptive and interpretive analysis.

C. Evaluation Dimensions

To ensure comparability across diverse studies, five evaluation dimensions are used consistently:

- *Demand forecasting accuracy*: how well the systems predict demand patterns and reduce mismatch between production and actual consumption.
- *Food waste reduction*: the extent to which interventions reduce food waste volumes and associated financial losses.
- *Supply chain and process transparency*: improvements in traceability coverage, data availability, and responsiveness to quality or safety issues.
- *Consumer or passenger satisfaction*: changes in convenience, perceived quality, reliability, and overall experience as reported in the studies.
- *Environmental impact*: documented effects on emissions or other environmental indicators, often measured through life cycle assessment or related methods.

These dimensions align with the research focus on operational performance, user experience, and sustainability, and they are used as lenses to interpret each article's contributions.

D. Data Analysis Approach

The analysis proceeds in three main steps:

- *Descriptive synthesis*: key features, methods, and results from each article are summarised in structured form, highlighting what was done, where, and with what outcomes.
- *Thematic analysis*: qualitative findings across studies are grouped into recurring themes, such as traceability challenges, forecasting benefits, organizational barriers, equity considerations, and circular economy opportunities.
- *Comparative interpretation*: the studies are compared along the five evaluation dimensions to identify typical performance ranges, trade-offs between technology-focused and organization-focused approaches, and gaps that are particularly relevant for railway food services.

This mixed-method analysis does not calculate new statistics but instead reinterprets existing quantitative and qualitative findings to produce an integrated, railway-oriented perspective on smart food ordering and waste reduction.

IV. Proposed Research Framework

A. Analytical Perspective for Railway Context

Because this paper does not implement a new system, the "framework" refers to an analytical lens rather than a concrete technical architecture. It specifies how existing systems and results are interpreted in relation to railway food services.

Two layers are emphasised:

- *Digital systems layer*: existing solutions for traceability, ML forecasting, IoT monitoring, smart kitchen waste tracking, and circular economy models, as reported in institutional and supply chain studies.
- *Service context layer*: the specific characteristics of railway services, including variable passenger demand, limited onboard facilities, tight schedules, and multi-station logistics.

The framework asks which patterns and limitations observed in non-rail contexts are likely to remain relevant if similar approaches are later adapted for trains.

B. Comparative Evaluation Dimensions

The same five evaluation dimensions from the methodology are used to structure the framework:

- Demand forecasting accuracy.
- Food waste reduction and cost impact.
- Supply chain and process transparency.
- Consumer or passenger satisfaction.
- Environmental impact and sustainability.

For each dimension, the reviewed articles provide examples of achievable improvements and common constraints, which are treated as benchmarks and cautionary signals for future railway-focused studies.

C. Phased Insight Generation

Instead of implementation phases, the framework suggests phases of research insight generation:

- *Phase 1: Evidence mapping* – summarise existing systems and results, as done in this article.
- *Phase 2: Contextual translation* – assess which technologies and outcomes appear most applicable to railway operations.
- *Phase 3: Scenario development* – outline hypothetical railway scenarios using performance ranges drawn from current literature.

- *Phase 4: Future study design* – propose empirical railway research or pilots informed by the identified gaps and constraints.

These phases are conceptual and oriented towards planning future mixed-method research, not towards describing a completed deployment.

V. Article Review and Analysis

A. Main Problems and Solutions

Across the ten papers, recurring issues and responses include:

- *Traceability gaps*: addressed with integrated systems combining QR/RFID, IoT, blockchain, and AI on cloud platforms.
- *Uncertain demand*: tackled by ML models such as LSTM linked to point-of-sale and context data to reduce overproduction in institutional settings.
- *Platform quality*: studied through consumer priorities (convenience, quality, price) and improved via better service and interface design in online delivery.
- *Cold chain risks*: reduced using IoT monitoring, predictive quality models, and mobile alerts in cold-chain logistics.
- *Fragmented transformation*: mitigated with prevention-first strategies, digital literacy training, and inclusive participation in institutional food services.
- *Reactive waste tracking*: improved by combining smart kitchen scales with feedback, workshops, and gamification.
- *Scaling circular solutions*: supported through regional networks, shared knowledge, and joint business models for waste valorisation.
- *Blockchain challenges*: managed via careful design around scalability, privacy, and regulation in agricultural supply chains.
- *Cost–sustainability split*: bridged through cost–waste analytics and triple-bottom-line reporting.
- *Weak environmental metrics*: addressed with standardised assessment approaches and better integration of Scope 3 emissions.

B. Technology vs. Organization Focus

Table I contrasts technology-focused and organization-focused interventions reported in the literature.

Aspect	**Technology-Focused**	**Organization-Focused**
Main mechanism	Automation and analytics	Training and culture change
Typical tools	Blockchain, IoT, AI/ML	Workshops, nudges, co-design
Cost level	Higher capital costs	Lower capital, staff time
Expertise needs	IT and data skills	Management and facilitation
Short-term effect	Fast visibility and control	Gradual behaviour shifts
Long-term stability	Risk if buy-in is weak	Stronger cultural embedding
Key limitation	Complexity and access	Limited without data systems

TABLE I
TECHNOLOGY- VS. ORGANIZATION-FOCUSED INTERVENTIONS.

The reviewed literature suggests that the strongest and most sustainable results appear when digital tools and organizational change measures are implemented together, even though most reported projects still focus more on one side than the other.

VI. Result Analysis and Discussion

A. Survey Overview

A primary survey titled "Train Food Service Survey: Passenger Preferences and Feedback" was conducted with 27 respondents to complement the literature review with empirical passenger data. The survey covered five sections: Demographics and Travel Habits, Current Train Food Experience, Mobile Ordering Preferences, Food Waste and Sustainability, and Transparency and Trust. Results are discussed below and mapped to the five evaluation dimensions of the research framework.

B. Demographics and Travel Habits

The respondent base is dominated by passengers aged 18–25 (88.9%), with minor representation from the 46–60 and 60+ brackets. Travel frequency shows 40.7% travel rarely by train and 33.3% several times a year. Regarding journey duration, 40.7% of respondents typically travel for more than 6 hours and 29.6% for 3–6 hours, meaning over 70% are on journeys long enough to require reliable on-board meal access.

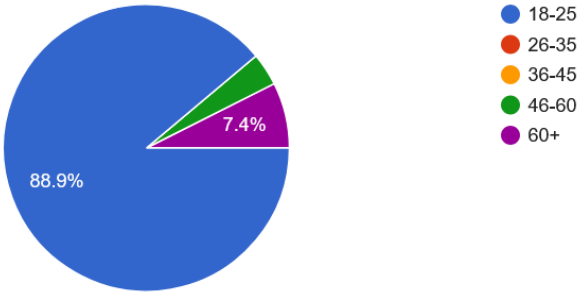

Fig. 1. Age group distribution of survey respondents (n=27).

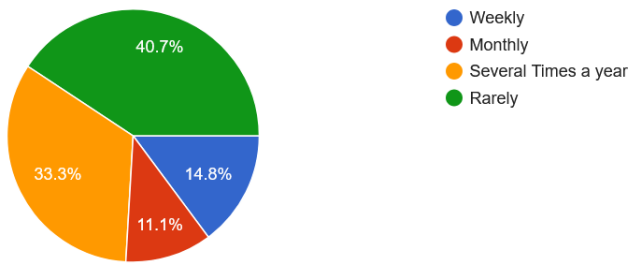

Fig. 2. Typical journey duration reported by respondents.

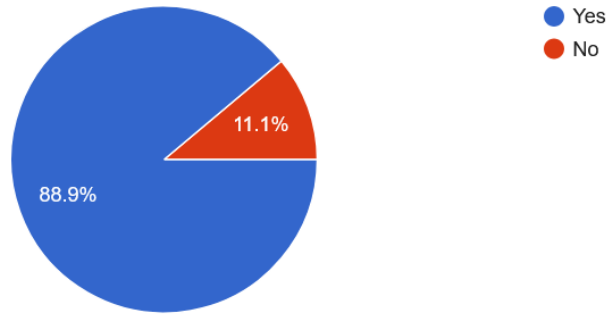

Fig. 4. Responses to "Have you ever seen food wasted on trains?"

C. Current Train Food Experience

Passenger satisfaction with existing train food services is critically low, with an average rating of 2.07 out of 5. Specifically, 44.4% gave the lowest rating (1) and only 7.4% gave the highest (5). Food purchase frequency indicates that 48.1% buy food on trains always or often, yet satisfaction remains very low, confirming a significant quality gap. Open-ended responses highlight common complaints: poor hygiene, substandard and overpriced food, and overall unreliability of the service. Most critically, 88.9% of respondents reported having personally observed food wasted on trains. This near-universal waste observation provides direct empirical grounding for the waste reduction focus of this paper [2].

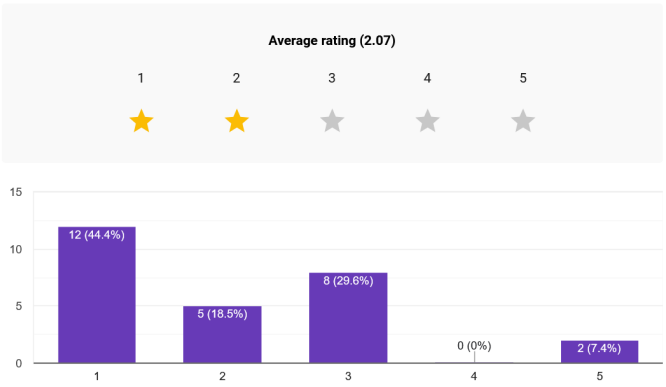

Fig. 3. Passenger satisfaction with current train food services (scale 1–5, average = 2.07).

D. Mobile Ordering Preferences

Approximately 51.8% of respondents expressed positive intent toward using a mobile app to order food on trains (33.3% "Definitely yes," 18.5% "Probably yes"), while 33.3% opposed and 14.8% were undecided. Preferred ordering timing was distributed across booking (18.5%), a few hours before the journey (29.6%), on-demand (29.6%), and after boarding (22.2%), indicating that a flexible multi-window ordering system is necessary rather than a rigid pre-booking model.

The most desired app features were: pre-ordering before the journey (63%), real-time order tracking (51.9%), ability to rate or review food (51.9%), multiple payment options (48.1%), and personalised recommendations (44.4%). Additionally, 77.8% of respondents expressed a desire for food recommendations based on past orders, confirming strong preference for a personalized user experience [3].

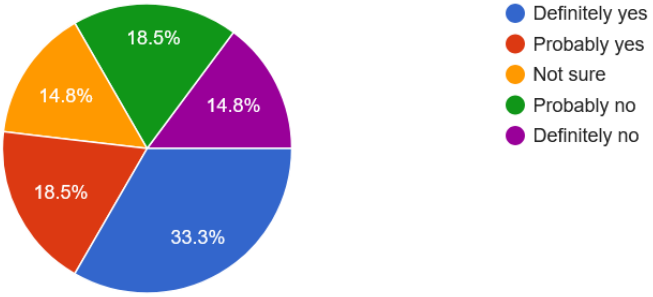

Fig. 5. Responses to "Would you use a mobile app to order food on trains?"

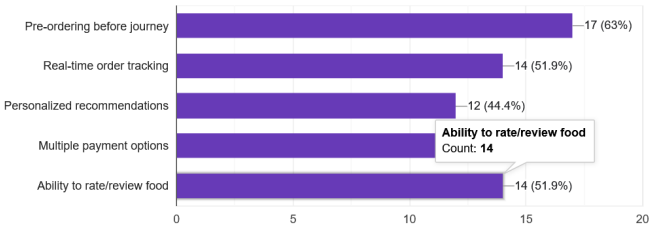

Fig. 6. Features that would encourage respondents to use a train food ordering app.

E. Food Waste and Sustainability

Concern about food waste on trains is high: 37% gave the maximum concern rating (5) and 29.6% rated it at 4, meaning 66.6% are highly concerned. Eco-friendly packaging was rated important by a clear majority, with 44.4% assigning it the highest importance score.

Pre-ordering to reduce waste was supported by 59.3% ("Yes") and conditionally by 29.6% ("Depends on convenience"), meaning approximately 89% would consider pre-ordering under suitable conditions. Since pre-ordering converts uncertain demand into committed orders before production begins, this finding directly supports the demand forecasting and waste reduction mechanisms identified across the reviewed studies [2], [5]. Dynamic pricing for early ordering was supported by 59.2% of respondents, consistent with cost–waste bridging strategies discussed in sustainability literature [9].

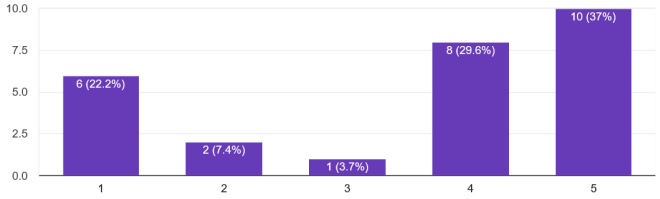

Fig. 7. Level of concern about food waste on trains (scale 1–5).

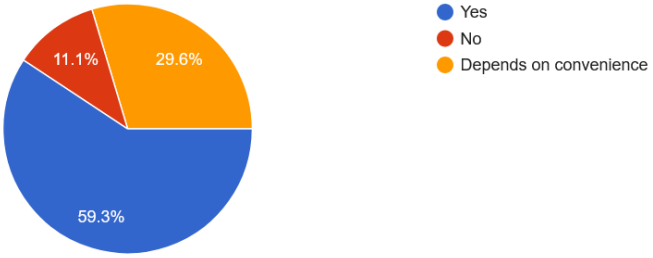

Fig. 8. Willingness to pre-order food to help reduce waste.

F. Transparency and Trust

Transparency preferences were the strongest signal in the survey. When asked whether they would like to see where their food comes from (kitchen location, preparation date), 77.8% responded "Yes." QR code scanning to view food origin and freshness was positively received by 70.3% of respondents (40.7% "Definitely yes," 29.6% "Probably yes"). Real-time food quality tracking via temperature sensors was rated at 4 or 5 out of 5 by 70.3% of respondents. These three findings collectively provide strong empirical support for including digital traceability and IoT-based quality monitoring in a future railway food ordering system [1], [4].

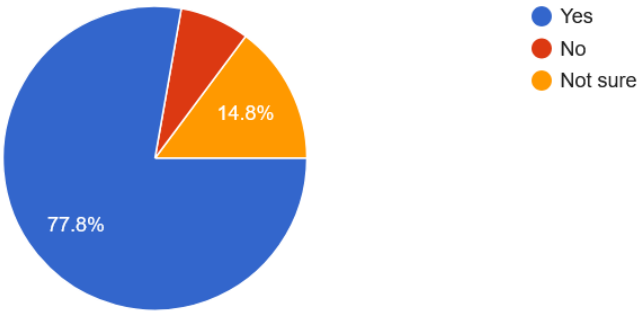

Fig. 9. Responses to "Would you like to see where your food comes from?"

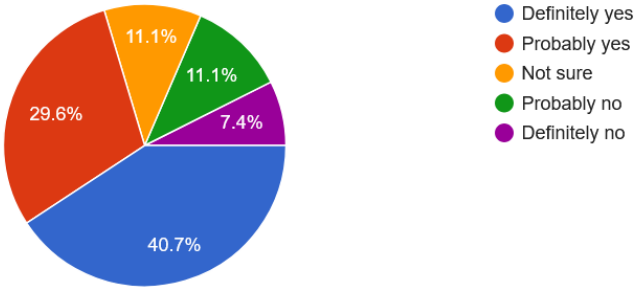

Fig. 10. Willingness to use QR code scanning to view food origin and freshness.

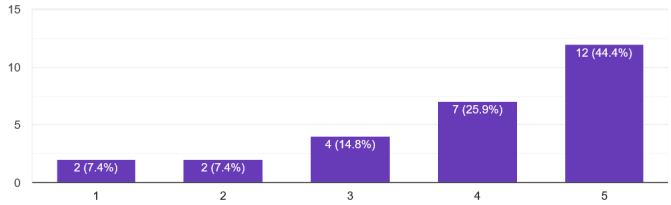

Fig. 11. Importance of real-time food quality tracking (e.g., temperature sensors), scale 1–5.

G. Discussion

The survey results validate and reinforce the five evaluation dimensions of the research framework. Table II maps the key findings to these dimensions.

The low average satisfaction (2.07/5) and the near-universal observation of food waste (88.9%) confirm that the problem

Finding	(%)	Dimension
Observed food waste on trains	88.9	Waste reduction
Pre-order support (yes/conditional)	88.9	Demand forecasting
Want food origin info	77.8	Traceability
QR code adoption intent	70.3	Traceability
IoT quality tracking (rated 4–5)	70.3	IoT monitoring
Dynamic pricing support	59.2	Cost–waste bridging
Mobile app adoption intent	51.8	Consumer satisfaction

TABLE II
SURVEY FINDINGS MAPPED TO RESEARCH FRAMEWORK DIMENSIONS.

is real and pressing. The strong interest in pre-ordering and dynamic pricing supports demand-side forecasting solutions consistent with the literature [2]. The high demand for transparency features validates the traceability and IoT components identified in reviewed studies [1], [4].

However, the survey also reveals a tension: while sustainability awareness and transparency preferences are high, mobile app adoption intent reaches only 51.8%. This gap between attitude and intended behaviour is consistent with findings on consumer barriers in online food delivery [3] and reinforces the argument that technology-focused solutions must be paired with organizational and usability measures, as summarised in Table I.

VII. CONCLUSION

This paper set out to integrate insights from existing digital food service research with empirical evidence from passengers to inform the design of a smart food ordering and waste reduction system for train services, without implementing a new system in practice. The mixed-method approach combined a structured review of ten articles with a primary survey of Bangladeshi train passengers, interpreted through five evaluation dimensions: demand forecasting, food waste reduction, transparency, consumer satisfaction, and environmental impact.

The literature review shows that ML-based demand forecasting, smart kitchen waste tracking, and cost–waste analytics can reduce overproduction and improve planning in institutional contexts, while IoT monitoring and blockchain traceability enhance visibility and responsiveness along food supply chains [2], [1], [5], [6], [7], [9]. At the same time, qualitative findings emphasise that technology-only interventions are rarely sufficient; sustained waste reduction and service improvements require complementary organizational measures such as staff training, co-design, and prevention-first strategies [5], [7].

The passenger survey confirms that similar problems and opportunities exist in the railway context. Satisfaction with current train food services is very low (average 2.07/5), and 88.9% of respondents reported personally observing food waste on trains, indicating a real and pressing operational and sustainability gap. At the same time, there is strong support for pre-ordering, dynamic pricing, eco-friendly packaging, and transparency features such as origin information, QR codes, and IoT-based quality tracking, which map directly onto the waste reduction, demand forecasting, traceability, and IoT monitoring dimensions of the framework.

However, the results also highlight a critical tension: sustainability concern and transparency preferences are high, yet mobile app adoption intent is only 51.8%. This attitude–behaviour gap mirrors barriers identified in online food delivery studies [3] and underscores that a railway food ordering solution must be not only technically robust but also usable, trustworthy, and aligned with passengers' routines. The comparison of technology-focused and organization-focused interventions suggests that combining digital tools with service design, staff practices, and communication strategies is likely to be more effective than deploying any single technology in isolation.

Taken together, these findings provide evidence-based expectations and design considerations for future railway-focused work. They suggest that promising system concepts will integrate pre-ordering and demand forecasting with dynamic pricing, digital traceability, and IoT quality monitoring, while embedding these capabilities within organizational processes that prioritise hygiene, reliability, and clear communication. Future research should therefore move from conceptual scenarios to pilot implementations and mixed-method evaluations in real railway settings, using the five evaluation dimensions as a common benchmark and explicitly testing how technical features and organizational measures interact to reduce waste and improve passenger experience over time.

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REFERENCES

[1] Ç. Tarhan and B. Gül, "Digital traceability in food supply chains: Technologies, benefits, and implementation challenges," *J. Food Supply Chain Manag.*, vol. 12, no. 3, pp. 245–267, 2025.
[2] S. Mattila, "Impacts of data-driven demand forecasting in reducing food waste and CO_2 emissions in campus restaurants: A case study," *Sustainability in Food Services*, vol. 8, no. 2, pp. 112–134, 2025.
[3] E. S. Varsini and K. M. Ranjani, "Decoding consumer behavior in online food delivery: A qualitative analysis," *J. Digital Consumer Res.*, vol. 15, no. 1, pp. 78–95, 2025.
[4] "IoT-based real-time monitoring in cold chain food supply," *World J. Adv. Res. Rev.*, vol. 19, no. 2, pp. 156–178, 2025.
[5] "Food waste reduction and digital transformation in institutional food services," *Int. J. Appl. Manag.*, vol. 22, no. 4, pp. 301–328, 2025.
[6] "Smart kitchen systems and waste monitoring technologies," *Food Service Technol. Rep.*, vol. 7, no. 3, pp. 189–215, 2023–2025.
[7] "Circular economy and sustainability in food service operations," *Eur. Sustain. Initiatives*, vol. 11, no. 2, pp. 234–261, 2024–2025.
[8] "Blockchain applications in agricultural supply chains," *World J. Adv. Res. Rev.*, vol. 18, no. 4, pp. 412–438, 2024.

[9] "Mitigating food waste: A new approach to sustainability and cost savings," *Business Sustain. Rev.*, vol. 14, no. 1, pp. 67–89, 2025.
[10] "Food waste and environmental management: Holistic assessment approaches," *J. Environ. Manag.*, vol. 28, no. 3, pp. 345–372, 2025.